

1 $\lambda_E^{\leq}$ (Seal)

1.1 Syntax

$$\begin{aligned}
A, B, C, R, S, \Gamma &::= \text{Int} \mid \varepsilon \mid A \rightarrow B \mid A \& B \mid \{\ell : A\} \mid \alpha_n \\
\Delta &\in \mathbb{N} \rightarrow \text{Typ} \quad \emptyset = \lambda n. \text{none} \\
e &::= ? \mid e.k \mid i \mid \varepsilon \mid \lambda(A \rightarrow B).e \mid e_1 \triangleright e_2 \mid \langle e_1, \lambda(A \rightarrow B).e_2 \rangle \mid e_1 e_2 \\
&\quad \mid e_1 \# e_2 \mid \{\ell = e\} \mid e.\ell \mid (e : A) \mid \langle n \rangle e \mid \text{seal } n \ R \ S \ e \mid \text{unseal } n \ R \ S \ e \\
v &::= i \mid \varepsilon \mid \langle v, \lambda(A \rightarrow B).e \rangle \mid \{\ell = v\} \mid v_1 \# v_2 \mid \langle n \rangle v
\end{aligned}$$

$$\begin{aligned}
\text{Int}[n:=R] &= \text{Int} \quad \varepsilon[n:=R] = \varepsilon \quad (A \rightarrow B)[n:=R] = A[n:=R] \rightarrow B[n:=R] \\
(A \& B)[n:=R] &= A[n:=R] \& B[n:=R] \quad \{\ell : A\}[n:=R] = \{\ell : A[n:=R]\} \quad \alpha_m[n:=R] = \begin{cases} R & m = n \\ \alpha_m & m \neq n \end{cases}
\end{aligned}$$

$$\begin{aligned}
&\frac{}{A \& B \ni_0 B}^{\text{ZERO}} \quad \frac{A \ni_k C}{A \& B \ni_{k+1} C}^{\text{SUCC}} \quad \frac{}{\ell \in \{\ell : A\}}^{\text{RCD}} \quad \frac{\ell \in A}{\ell \in A \& B}^{\text{ANDL}} \quad \frac{\ell \in B}{\ell \in A \& B}^{\text{ANDR}} \\
&\frac{}{\ell : A \in \{\ell : A\}}^{\text{ZERO}} \quad \frac{\ell : C \in A \quad \neg \ell \in B}{\ell : C \in A \& B}^{\text{LANDL}} \quad \frac{\ell : C \in B \quad \neg \ell \in A}{\ell : C \in A \& B}^{\text{LANDR}} \\
&\frac{}{\text{Int ord}}^{\text{OINT}} \quad \frac{}{A \rightarrow B \text{ ord}}^{\text{OARR}} \quad \frac{}{\{\ell : A\} \text{ ord}}^{\text{ORCD}} \quad \frac{}{\alpha_n \text{ ord}}^{\text{OBRAND}} \quad \frac{}{\lceil \varepsilon \rceil}^{\text{TLTOP}} \\
&\frac{\lceil A \rceil \quad \lceil B \rceil}{\lceil A \& B \rceil}^{\text{TLAND}} \quad \frac{\lceil B \rceil}{\lceil A \rightarrow B \rceil}^{\text{TLARR}} \quad \frac{\lceil B \rceil}{\lceil \{\ell : B\} \rceil}^{\text{TLRCD}}
\end{aligned}$$

$$\text{Int}^\dagger = \varepsilon \quad \varepsilon^\dagger = \varepsilon \quad (A \rightarrow B)^\dagger = \langle \varepsilon, \lambda(A \rightarrow B).B^\dagger \rangle \quad (A \& B)^\dagger = A^\dagger \# B^\dagger \quad \{\ell : A\}^\dagger = \{\ell = A^\dagger\} \quad \alpha_n^\dagger = \varepsilon$$

1.2 Typing

Subtyping $A <: B$

$$\begin{aligned}
&\frac{}{\text{Int} <: \text{Int}}^{\text{SINT}} \quad \frac{}{A <: \varepsilon}^{\text{STOP}} \quad \frac{B_1 <: A_1 \quad A_2 <: B_2}{A_1 \rightarrow A_2 <: B_1 \rightarrow B_2}^{\text{SARR}} \quad \frac{A_1 <: A_3}{A_1 \& A_2 <: A_3}^{\text{SANDL}} \\
&\frac{A_2 <: A_3}{A_1 \& A_2 <: A_3}^{\text{SANDR}} \quad \frac{A_1 <: A_2 \quad A_1 <: A_3}{A_1 <: A_2 \& A_3}^{\text{SAND}} \quad \frac{A <: B}{\{\ell : A\} <: \{\ell : B\}}^{\text{SRCD}} \quad \frac{}{\alpha_n <: \alpha_n}^{\text{SBRAND}}
\end{aligned}$$

Common ordinary supertypes and disjointness $A \sqcap B, \quad A * B := \neg(A \sqcap B)$

$$\begin{aligned}
&\frac{}{\text{Int} \sqcap \text{Int}}^{\text{CINT}} \quad \frac{A \sqcap C}{A \& B \sqcap C}^{\text{CANDL}} \quad \frac{B \sqcap C}{A \& B \sqcap C}^{\text{CANDR}} \quad \frac{A \sqcap B}{A \sqcap B \& C}^{\text{CRANDL}} \quad \frac{A \sqcap C}{A \sqcap B \& C}^{\text{CRANDR}} \\
&\frac{B \sqcap D}{A \rightarrow B \sqcap C \rightarrow D}^{\text{CARR}} \quad \frac{A \sqcap B}{\{\ell : A\} \sqcap \{\ell : B\}}^{\text{CRCD}} \quad \frac{}{\alpha_n \sqcap \alpha_n}^{\text{CBRAND}}
\end{aligned}$$

Casting and consistency $v \hookrightarrow_A v', \quad v_1 \approx v_2 := \forall A w_1 w_2. v_1 \hookrightarrow_A w_1 \Rightarrow v_2 \hookrightarrow_A w_2 \Rightarrow w_1 = w_2$

$$\begin{array}{c}
\frac{}{i \hookrightarrow_{\text{Int}} i}^{\text{CINT}} \quad \frac{}{v \hookrightarrow_{\varepsilon} \varepsilon}^{\text{CTOP}} \quad \frac{\neg \text{]}D[\quad C <: A \quad B <: D}{\langle v, \lambda(A \rightarrow B).e \rangle \hookrightarrow_{C \rightarrow D} \langle v, \lambda(A \rightarrow D).e \rangle}^{\text{CARROW}} \\
\\
\frac{\text{]}D[\quad C <: A \quad B <: D}{\langle v, \lambda(A \rightarrow B).e \rangle \hookrightarrow_{C \rightarrow D} \langle \varepsilon, \lambda(C \rightarrow D).D^\uparrow \rangle}^{\text{CARROWTL}} \quad \frac{A \text{ ord} \quad v_1 \hookrightarrow_A v'_1}{v_1 \# v_2 \hookrightarrow_A v'_1}^{\text{CMRGL}} \\
\\
\frac{A \text{ ord} \quad v_2 \hookrightarrow_A v'_2}{v_1 \# v_2 \hookrightarrow_A v'_2}^{\text{CMRGR}} \quad \frac{v \hookrightarrow_A v_1 \quad v \hookrightarrow_B v_2}{v \hookrightarrow_{A \& B} v_1 \# v_2}^{\text{CAND}} \quad \frac{v \hookrightarrow_A v'}{\{\ell = v\} \hookrightarrow_{\{\ell:A\}} \{\ell = v'\}}^{\text{CRCD}} \\
\\
\frac{}{\langle n \rangle v \hookrightarrow_{\alpha_n} \langle n \rangle v}^{\text{CWRAP}}
\end{array}$$

Reserved-label freedom and signature well-formedness $\text{NoRes } T, \quad \text{WfSig } n \ R \ S$

$$\begin{array}{c}
\frac{}{\text{NoRes Int}} \quad \frac{}{\text{NoRes } \varepsilon} \quad \frac{}{\text{NoRes } \alpha_n} \quad \frac{\text{NoRes } A \quad \text{NoRes } B}{\text{NoRes } (A \rightarrow B)} \quad \frac{\ell \neq \#f \quad \text{NoRes } A}{\text{NoRes } \{\ell : A\}} \\
\\
\frac{\text{NoRes } A \quad \text{NoRes } B}{\text{NoRes } (A \& B)} \\
\\
\frac{}{\text{WfSig } n \ R \ \text{Int}} \quad \frac{}{\text{WfSig } n \ R \ \varepsilon} \quad \frac{}{\text{WfSig } n \ R \ \alpha_m} \quad \frac{\text{WfSig } n \ R \ A \quad \text{WfSig } n \ R \ B}{\text{WfSig } n \ R \ (A \rightarrow B)} \\
\\
\frac{\ell \neq \#f \quad \text{WfSig } n \ R \ A}{\text{WfSig } n \ R \ \{\ell : A\}} \quad \frac{\text{WfSig } n \ R \ A \quad \text{WfSig } n \ R \ B \quad A * B \quad A[n:=R] * B[n:=R]}{\text{WfSig } n \ R \ (A \& B)}
\end{array}$$

Typing $\Delta; \Gamma \vdash e : A$

$$\begin{array}{c}
\frac{}{\Delta; \Gamma \vdash ? : \Gamma} \text{TQUERY} \quad \frac{}{\Delta; \Gamma \vdash i : \text{Int}} \text{TINT} \quad \frac{}{\Delta; \Gamma \vdash \varepsilon : \varepsilon} \text{TUNIT} \quad \frac{\Delta; \Gamma \vdash e_1 : A \rightarrow B \quad \Delta; \Gamma \vdash e_2 : A}{\Delta; \Gamma \vdash e_1 e_2 : B} \text{TAPP} \\
\\
\frac{\Delta; \Gamma \vdash e_1 : \Gamma_1 \quad \Delta; \Gamma_1 \vdash e_2 : A}{\Delta; \Gamma \vdash e_1 \triangleright e_2 : A} \text{TBOX} \quad \frac{\Delta; \Gamma \vdash e : B \quad B \ni_k A}{\Delta; \Gamma \vdash e.k : A} \text{TPROJ} \quad \frac{\Delta; \Gamma \vdash e : A}{\Delta; \Gamma \vdash \{\ell = e\} : \{\ell : A\}} \text{TRCD} \\
\\
\frac{\Delta; \Gamma \vdash e : B \quad \ell : A \in B}{\Delta; \Gamma \vdash e.\ell : A} \text{TRPROJ} \quad \frac{\Delta; \Gamma \vdash e_1 : A \quad \Delta; \Gamma \& A \vdash e_2 : B \quad A * \Gamma \quad A * B}{\Delta; \Gamma \vdash e_1 \# e_2 : A \& B} \text{TMRG} \\
\\
\frac{v_1 \text{ value} \quad v_2 \text{ value} \quad \Delta; \varepsilon \vdash v_1 : A \quad \Delta; \varepsilon \vdash v_2 : B \quad v_1 \approx v_2}{\Delta; \Gamma \vdash v_1 \# v_2 : A \& B} \text{TMERGEV} \\
\\
\frac{\Gamma * A \quad \Delta; \Gamma \& A \vdash e : B}{\Delta; \Gamma \vdash \lambda(A \rightarrow B).e : A \rightarrow B} \text{TLAM} \\
\\
\frac{v \text{ value} \quad \Delta; \varepsilon \vdash v : \Gamma_1 \quad \Gamma_1 * A \quad \Delta; \Gamma_1 \& A \vdash e : B' \quad B' <: B \quad C <: A}{\Delta; \Gamma \vdash \langle v, \lambda(A \rightarrow B).e \rangle : C \rightarrow B} \text{TCLOS} \\
\\
\frac{\Delta; \Gamma \vdash e : B \quad B <: A}{\Delta; \Gamma \vdash (e : A) : A} \text{TANNO} \quad \frac{\Delta n = R \quad v \text{ value} \quad \Delta; \varepsilon \vdash v : R}{\Delta; \Gamma \vdash \langle n \rangle v : \alpha_n} \text{TWRAP} \\
\\
\frac{\Delta n = R \quad \text{NoRes } R \quad \text{WfSig } n \ R \ S \quad \Delta; \Gamma \vdash e : S[n := R]}{\Delta; \Gamma \vdash \text{seal } n \ R \ S \ e : S} \text{TSEAL} \\
\\
\frac{\Delta n = R \quad \text{NoRes } R \quad \text{WfSig } n \ R \ S \quad \Delta; \Gamma \vdash e : S}{\Delta; \Gamma \vdash \text{unseal } n \ R \ S \ e : S[n := R]} \text{TUNSEAL}
\end{array}$$

$$\Delta \sqsubseteq \Delta' := \forall n R. \Delta n = R \Rightarrow \Delta' n = R$$

1.3 Semantics

Value lookups $v \ni_k v', \ell : v' \in v$

$$\begin{array}{c}
\frac{}{v_1 \# v_2 \ni_0 v_2} \text{LVZERO} \quad \frac{v_1 \ni_k v_3}{v_1 \# v_2 \ni_{k+1} v_3} \text{LVSUCC} \quad \frac{}{\ell : e \in \{\ell = e\}} \text{RVLZERO} \quad \frac{\ell : e \in e_1}{\ell : e \in e_1 \# e_2} \text{VLANDL} \\
\\
\frac{\ell : e \in e_2}{\ell : e \in e_1 \# e_2} \text{VLANDR}
\end{array}$$

Sealing coercions $v \hookrightarrow_S^{\text{seal } n \ R} w, \quad v \hookrightarrow_S^{\text{unseal } n \ R} w \quad (\text{pxe}(c) := \varepsilon \# \{\#f = c\}, \quad \text{pxf} := (? . 1). \#f)$

$$\begin{array}{c}
\frac{}{v \hookrightarrow_{\alpha_n}^{\text{seal } n \ R} \langle n \rangle v} \text{BRAND-EQ} \quad \frac{m \neq n}{v \hookrightarrow_{\alpha_m}^{\text{seal } n \ R} v} \text{BRAND-NE} \quad \frac{}{i \hookrightarrow_{\text{Int}}^{\text{seal } n \ R} i} \text{INT} \quad \frac{}{v \hookrightarrow_{\varepsilon}^{\text{seal } n \ R} \varepsilon} \text{TOP} \\
\\
\frac{v_1 \hookrightarrow_A^{\text{seal } n \ R} w_1 \quad v_2 \hookrightarrow_B^{\text{seal } n \ R} w_2}{v_1 \# v_2 \hookrightarrow_{A \& B}^{\text{seal } n \ R} w_1 \# w_2} \text{AND} \quad \frac{v \hookrightarrow_A^{\text{seal } n \ R} w}{\{\ell = v\} \hookrightarrow_{\{\ell : A\}}^{\text{seal } n \ R} \{\ell = w\}} \text{RCD} \\
\\
\frac{}{c \hookrightarrow_{A \rightarrow B}^{\text{seal } n \ R} \langle \text{pxe}(c), \lambda(A \rightarrow B). \text{seal } n \ R \ B \ (\text{pxf} (\text{unseal } n \ R \ A \ ? . 0)) \rangle} \text{ARR}
\end{array}$$

$$\begin{array}{c}
\frac{}{\langle n \rangle v \hookrightarrow_{\alpha_n}^{\text{unseal } n R} v} \text{ BRAND-EQ} \quad \frac{m \neq n}{v \hookrightarrow_{\alpha_m}^{\text{unseal } n R} v} \text{ BRAND-NE} \quad \frac{}{i \hookrightarrow_{\text{Int}}^{\text{unseal } n R} i} \text{ INT} \quad \frac{}{v \hookrightarrow_{\varepsilon}^{\text{unseal } n R} \varepsilon} \text{ TOP} \\
\\
\frac{v_1 \hookrightarrow_A^{\text{unseal } n R} w_1 \quad v_2 \hookrightarrow_B^{\text{unseal } n R} w_2}{v_1 \# v_2 \hookrightarrow_{A \& B}^{\text{unseal } n R} w_1 \# w_2} \text{ AND} \quad \frac{v \hookrightarrow_A^{\text{unseal } n R} w}{\{\ell = v\} \hookrightarrow_{\{\ell:A\}}^{\text{unseal } n R} \{\ell = w\}} \text{ RCD} \\
\\
\frac{}{c \hookrightarrow_{A \rightarrow B}^{\text{unseal } n R} \langle \text{pxe}(c), \lambda(A[n:=R] \rightarrow B[n:=R]). \text{unseal } n R B (\text{pxf} (\text{seal } n R A ? . 0)) \rangle} \text{ ARR}
\end{array}$$

Small-step $v \vdash e \hookrightarrow e'$ (all rules carry v value; value premises shown)

$$\begin{array}{c}
\frac{}{v \vdash ? \hookrightarrow v} \text{ SQUERY} \quad \frac{v \vdash e_1 \hookrightarrow e'_1}{v \vdash e_1 e_2 \hookrightarrow e'_1 e_2} \text{ SAPPL} \quad \frac{v_1 \text{ value} \quad v \vdash e_2 \hookrightarrow e'_2}{v \vdash v_1 e_2 \hookrightarrow v_1 e'_2} \text{ SAPPR} \\
\\
\frac{v_1 \text{ value} \quad v_2 \text{ value} \quad v_2 \hookrightarrow_A v'_2}{v \vdash \langle v_1, \lambda(A \rightarrow B).e \rangle v_2 \hookrightarrow (v_1 \# v'_2) \triangleright (e : B)} \text{ SBETA} \quad \frac{}{v \vdash \lambda(A \rightarrow B).e \hookrightarrow \langle v, \lambda(A \rightarrow B).e \rangle} \text{ SCLOS} \\
\\
\frac{v \vdash e_1 \hookrightarrow e'_1}{v \vdash e_1 \triangleright e_2 \hookrightarrow e'_1 \triangleright e_2} \text{ SBOXL} \quad \frac{v_1 \text{ value} \quad v_1 \vdash e_2 \hookrightarrow e'_2}{v \vdash v_1 \triangleright e_2 \hookrightarrow v_1 \triangleright e'_2} \text{ SBOXR} \quad \frac{v_1 \text{ value} \quad v_2 \text{ value}}{v \vdash v_1 \triangleright v_2 \hookrightarrow v_2} \text{ SBOXV} \\
\\
\frac{v \vdash e_1 \hookrightarrow e'_1}{v \vdash e_1 \# e_2 \hookrightarrow e'_1 \# e_2} \text{ SMRGL} \quad \frac{v_1 \text{ value} \quad v \# v_1 \vdash e_2 \hookrightarrow e'_2}{v \vdash v_1 \# e_2 \hookrightarrow v_1 \# e'_2} \text{ SMRGR} \quad \frac{v \vdash e \hookrightarrow e'}{v \vdash e.k \hookrightarrow e'.k} \text{ SPROJ} \\
\\
\frac{v_1 \text{ value} \quad v_1 \ni_k v'}{v \vdash v_1.k \hookrightarrow v'} \text{ SPROJV} \quad \frac{v \vdash e \hookrightarrow e'}{v \vdash \{\ell = e\} \hookrightarrow \{\ell = e'\}} \text{ SLREC} \quad \frac{v \vdash e \hookrightarrow e'}{v \vdash e.\ell \hookrightarrow e'.\ell} \text{ SRPROJ} \\
\\
\frac{v_1 \text{ value} \quad \ell : v' \in v_1}{v \vdash v_1.\ell \hookrightarrow v'} \text{ SRPROJV} \quad \frac{v \vdash e \hookrightarrow e'}{v \vdash (e : A) \hookrightarrow (e' : A)} \text{ SANNO} \quad \frac{v_1 \text{ value} \quad v_1 \hookrightarrow_A v'}{v \vdash (v_1 : A) \hookrightarrow v'} \text{ SANN OV} \\
\\
\frac{v \vdash e \hookrightarrow e'}{v \vdash \langle n \rangle e \hookrightarrow \langle n \rangle e'} \text{ SWRAP} \quad \frac{v \vdash e \hookrightarrow e'}{v \vdash \text{seal } n R S e \hookrightarrow \text{seal } n R S e'} \text{ SSEAL} \quad \frac{v_1 \text{ value} \quad v_1 \hookrightarrow_S^{\text{seal } n R} w}{v \vdash \text{seal } n R S v_1 \hookrightarrow w} \text{ SSEALV} \\
\\
\frac{v \vdash e \hookrightarrow e'}{v \vdash \text{unseal } n R S e \hookrightarrow \text{unseal } n R S e'} \text{ SUNSEAL} \quad \frac{v_1 \text{ value} \quad v_1 \hookrightarrow_S^{\text{unseal } n R} w}{v \vdash \text{unseal } n R S v_1 \hookrightarrow w} \text{ SUNSEALV} \\
\\
\frac{}{v \vdash e \hookrightarrow^* e} \text{ REFL} \quad \frac{v \vdash e_1 \hookrightarrow e_2 \quad v \vdash e_2 \hookrightarrow^* e_3}{v \vdash e_1 \hookrightarrow^* e_3} \text{ STEP}
\end{array}$$

1.4 Meta-theory

Theorem 1.1 (gdeterminism).

$$\Delta; \Gamma \vdash e : A \wedge \Delta; \varepsilon \vdash v : \Gamma \wedge v \vdash e \hookrightarrow e_1 \wedge v \vdash e \hookrightarrow e_2 \implies e_1 = e_2$$

Theorem 1.2 (gprogress).

$$\Delta; \Gamma \vdash e : A \wedge v \text{ value} \wedge \Delta; \varepsilon \vdash v : \Gamma \implies e \text{ value} \vee \exists e'. v \vdash e \hookrightarrow e'$$

Theorem 1.3 (gpreservation).

$$v \vdash e \hookrightarrow e' \wedge \Delta; \Gamma \vdash e : A \wedge \Delta; \varepsilon \vdash v : \Gamma \implies \Delta; \Gamma \vdash e' : A$$

Theorem 1.4 (hastype_weaken_store).

$$\Delta \sqsubseteq \Delta' \wedge \Delta; \Gamma \vdash e : A \implies \Delta'; \Gamma \vdash e : A$$

Definition 1.5 (viewL, viewR, WrapRel, RawRel, LRg, LR). $o \in \{\text{none}\} \cup \mathbb{N} \times \text{Typ} \times \text{Typ}$; $\eta : \mathbb{N} \rightarrow \text{Exp} \rightarrow \text{Exp} \rightarrow \text{Prop}$; *stores* Δ_1, Δ_2 *ambient*.

$$\begin{aligned} \text{viewL } \text{none } T &= T & \text{viewL } (n, R_1, R_2) T &= T[n := R_1] \\ \text{viewR } \text{none } T &= T & \text{viewR } (n, R_1, R_2) T &= T[n := R_2] \end{aligned}$$

$$\begin{aligned} \text{WrapRel } m (v_1, v_2) &= \Delta_1; \varepsilon \vdash v_1 : \alpha_m \wedge \Delta_2; \varepsilon \vdash v_2 : \alpha_m \wedge \exists w_1 w_2. v_1 = \langle m \rangle w_1 \wedge v_2 = \langle m \rangle w_2 \wedge \eta m w_1 w_2 \\ \text{RawRel } n R_1 R_2 (v_1, v_2) &= v_1 \text{ value} \wedge v_2 \text{ value} \wedge \Delta_1; \varepsilon \vdash v_1 : R_1 \wedge \Delta_2; \varepsilon \vdash v_2 : R_2 \wedge \eta n v_1 v_2 \end{aligned}$$

$$\begin{aligned} \mathcal{V}[\text{Int}]^o(v_1, v_2) &= \exists i. v_1 = i \wedge v_2 = i \\ \mathcal{V}[\varepsilon]^o(v_1, v_2) &= v_1 = \varepsilon \wedge v_2 = \varepsilon \\ \mathcal{V}[\alpha_m]^o(v_1, v_2) &= \begin{cases} \text{RawRel } n R_1 R_2 (v_1, v_2) & o = (n, R_1, R_2), m = n \\ \text{WrapRel } m (v_1, v_2) & \text{otherwise} \end{cases} \\ \mathcal{V}[A \& B]^o(v_1, v_2) &= \Delta_1; \varepsilon \vdash v_1 : \text{viewL } o (A \& B) \wedge \Delta_2; \varepsilon \vdash v_2 : \text{viewR } o (A \& B) \wedge \\ &\quad \exists a_1 b_1 a_2 b_2. v_1 = a_1 \# b_1 \wedge v_2 = a_2 \# b_2 \wedge \mathcal{V}[A]^o(a_1, a_2) \wedge \mathcal{V}[B]^o(b_1, b_2) \\ \mathcal{V}[\{\ell : A\}]^o(v_1, v_2) &= \exists w_1 w_2. v_1 = \{\ell = w_1\} \wedge v_2 = \{\ell = w_2\} \wedge \mathcal{V}[A]^o(w_1, w_2) \\ \mathcal{V}[A \rightarrow B]^o(v_1, v_2) &= v_1 \text{ value} \wedge v_2 \text{ value} \wedge \Delta_1; \varepsilon \vdash v_1 : \text{viewL } o (A \rightarrow B) \wedge \Delta_2; \varepsilon \vdash v_2 : \text{viewR } o (A \rightarrow B) \wedge \\ &\quad \forall u_1 u_2. \mathcal{V}[A]^o(u_1, u_2) \Rightarrow \forall \rho_1 \rho_2. \rho_1 \text{ value} \Rightarrow \rho_2 \text{ value} \Rightarrow \\ &\quad \exists w_1 w_2. \rho_1 \vdash v_1 u_1 \hookrightarrow^* w_1 \wedge \rho_2 \vdash v_2 u_2 \hookrightarrow^* w_2 \wedge \mathcal{V}[B]^o(w_1, w_2) \\ \mathcal{V}[T] &:= \mathcal{V}[T]^{\text{none}} \end{aligned}$$

Definition 1.6 (OpenAdm, OpenBrand, SemTyp).

$$\begin{aligned} \text{OpenAdm } \eta o &= \forall n R_1 R_2. o = (n, R_1, R_2) \Rightarrow \neg \lceil R_1 \rceil \wedge \neg \lceil R_2 \rceil \wedge \\ &\quad (\forall w_1 w_2. \eta n w_1 w_2 \Rightarrow \forall w'_1 w'_2. w_1 \hookrightarrow_{R_1} w'_1 \Rightarrow w_2 \hookrightarrow_{R_2} w'_2 \Rightarrow \eta n w'_1 w'_2) \end{aligned}$$

$$\text{OpenBrand } \Delta_1 \Delta_2 \eta n R_1 R_2 = \begin{cases} \Delta_1 n = R_1 \wedge \Delta_2 n = R_2 \wedge \text{NoRes } R_1 \wedge \text{NoRes } R_2 \wedge \neg \lceil R_1 \rceil \wedge \neg \lceil R_2 \rceil \wedge \\ \forall w_1 w_2. \eta n w_1 w_2 \Rightarrow \forall w'_1 w'_2. w_1 \hookrightarrow_{R_1} w'_1 \Rightarrow w_2 \hookrightarrow_{R_2} w'_2 \Rightarrow \eta n w'_1 w'_2 \end{cases}$$

$$\text{SemTyp } \Delta_1 \Delta_2 \eta \Gamma e_1 e_2 A = \forall \rho_1 \rho_2. \mathcal{V}[\Gamma](\rho_1, \rho_2) \Rightarrow \exists w_1 w_2. \rho_1 \vdash e_1 \hookrightarrow^* w_1 \wedge \rho_2 \vdash e_2 \hookrightarrow^* w_2 \wedge \mathcal{V}[A](w_1, w_2)$$

Theorem 1.7 (fundamental).

$$\emptyset; \Gamma \vdash e : A \implies \text{SemTyp } \Delta_1 \Delta_2 \eta \Gamma e e A$$

Theorem 1.8 (normalization).

$$\emptyset; \varepsilon \vdash e : A \implies \exists v. v \text{ value} \wedge \varepsilon \vdash e \hookrightarrow^* v$$

Theorem 1.9 (sealing).

$$\mathcal{V}[A](p'_1, p'_2) \wedge \emptyset; A \vdash e : \text{Int} \implies \exists i. p'_1 \vdash e \hookrightarrow^* i \wedge p'_2 \vdash e \hookrightarrow^* i$$

Theorem 1.10 (coe_{lr}).

$$\text{OpenBrand } \Delta_1 \Delta_2 \eta n R_1 R_2 \wedge \text{WfSig } n R_1 S \wedge \text{WfSig } n R_2 S \implies$$

$$\begin{aligned} (\forall v_1 v_2 w_1 w_2. \mathcal{V}[S]^{(n, R_1, R_2)}(v_1, v_2) \Rightarrow v_1 \xrightarrow{S}^{\text{seal } n R_1} w_1 \Rightarrow v_2 \xrightarrow{S}^{\text{seal } n R_2} w_2 \Rightarrow \mathcal{V}[S](w_1, w_2)) \wedge \\ (\forall v_1 v_2 w_1 w_2. \mathcal{V}[S](v_1, v_2) \Rightarrow v_1 \xrightarrow{S}^{\text{unseal } n R_1} w_1 \Rightarrow v_2 \xrightarrow{S}^{\text{unseal } n R_2} w_2 \Rightarrow \mathcal{V}[S]^{(n, R_1, R_2)}(w_1, w_2)) \end{aligned}$$

Theorem 1.11 (representation_independence).

$$\begin{aligned} \text{OpenBrand } \Delta_1 \Delta_2 \eta n R_1 R_2 \wedge \text{WfSig } n R_1 S \wedge \text{WfSig } n R_2 S \wedge \mathcal{V}[S]^{(n, R_1, R_2)}(p_1, p_2) \wedge \emptyset; S \vdash e : \text{Int} \\ \implies \exists i. \varepsilon \vdash (\text{seal } n R_1 S p_1) \triangleright e \hookrightarrow^* i \wedge \varepsilon \vdash (\text{seal } n R_2 S p_2) \triangleright e \hookrightarrow^* i \end{aligned}$$

2 λ_{SCE} (SCE)

2.1 Syntax

$$\begin{aligned}
T, A, B, R, S, \Gamma &::= \text{Int} \mid \varepsilon \mid T \rightarrow T \mid T \& T \mid T \vee T \mid \{\ell : T\} \mid \text{sig } \sigma \mid X_k \mid \mu T \mid \alpha_n \\
\sigma &::= \langle T \rangle \mid T \Rightarrow \sigma \\
s &::= \bullet \mid \circ \quad \Delta \in \mathbb{N} \rightarrow \text{Typ} \quad \emptyset = \lambda n. \text{none} \quad \Delta \sqsubseteq \Delta' := \forall n. R. \Delta n = R \Rightarrow \Delta' n = R \\
e &::= ? \mid e.k \mid i \mid \varepsilon \mid \lambda A. e \mid e_1 \triangleright e_2 \mid \langle e_1, \lambda A. e_2 \rangle \mid e_1 e_2 \mid e_1 \# e_2 \mid \{\ell = e\} \mid e.\ell \\
&\quad \mid \text{struct}^s e \mid \text{functor}^s(A).e \mid \langle e_1, \text{functor}(A).e_2 \rangle \mid e_1 \ltimes e_2 \mid e_1 \bullet e_2 \mid e_1 \ltimes^* e_2 \\
&\quad \mid e_1 \text{ ,, } e_2 \mid \text{let}^A e_1 \text{ in } e_2 \mid \text{open } e_1 \text{ in } e_2 \\
&\quad \mid \text{inl}^B e \mid \text{inr}^A e \mid \text{case } e \text{ of } e_1 \mid e_2 \mid \mu \lambda(A \rightarrow B). e \mid \langle e_1, \mu \lambda(A \rightarrow B). e_2 \rangle \mid \text{fold}^T e \mid \text{unfold } e \\
&\quad \mid \langle n \rangle e \mid \text{seal } n \text{ } R \text{ } S \text{ } e \mid \text{unseal } n \text{ } R \text{ } S \text{ } e \\
v &::= i \mid \varepsilon \mid \langle v, \lambda A. e \rangle \mid \langle v, \text{functor}(A).e \rangle \mid v_1 \# v_2 \mid \{\ell = v\} \mid \text{inl}^B v \mid \text{inr}^A v \mid \langle v, \mu \lambda(A \rightarrow B). e \rangle \mid \text{fold}^T v \mid \langle n \rangle v
\end{aligned}$$

$$\begin{aligned}
X_k[d:=S]^\mu &= \begin{cases} S & k = d \\ X_k & k \neq d \end{cases} \quad (\mu T)[d:=S]^\mu = \mu (T[d+1:=S]^\mu) \quad \alpha_n[d:=S]^\mu = \alpha_n \quad (\text{homomorphic otherwise, incl. } \sigma) \\
\alpha_m[n:=R] &= \begin{cases} R & m = n \\ \alpha_m & m \neq n \end{cases} \quad X_k[n:=R] = X_k \quad (\mu T)[n:=R] = \mu (T[n:=R]) \quad (\text{homomorphic otherwise, incl. } \sigma)
\end{aligned}$$

$$\begin{aligned}
&\frac{}{A \& B \ni_0 B} \text{ZERO} \quad \frac{A \ni_k C}{A \& B \ni_{k+1} C} \text{SUCC} \quad \frac{}{\ell \in \{\ell : T\}} \text{RCD} \quad \frac{\ell \in A}{\ell \in A \& B} \text{ANDL} \quad \frac{\ell \in B}{\ell \in A \& B} \text{ANDR} \\
&\frac{\ell \in T}{\ell \in \text{sig } \langle T \rangle} \text{SIG} \quad \frac{}{\ell : T \in \{\ell : T\}} \text{ZERO} \quad \frac{\ell : T \in A \quad \neg \ell \in B}{\ell : T \in A \& B} \text{ANDL} \quad \frac{\ell : T \in B \quad \neg \ell \in A}{\ell : T \in A \& B} \text{ANDR} \\
&\frac{\ell : A \in T}{\ell : A \in \text{sig } \langle T \rangle} \text{SIG}
\end{aligned}$$

2.2 Elaboration (SCE $\rightarrow \lambda_E^{\leq}$)

Type translation $\llbracket \cdot \rrbracket : \text{SCE.Type} \rightarrow \text{Seal.Type}, \quad \llbracket \Delta \rrbracket := \lambda n. \llbracket \Delta n \rrbracket$

$$\begin{aligned}
\llbracket \text{Int} \rrbracket &= \text{Int} \quad \llbracket \varepsilon \rrbracket = \varepsilon \quad \llbracket A \rightarrow B \rrbracket = \llbracket A \rrbracket \rightarrow \llbracket B \rrbracket \quad \llbracket A \& B \rrbracket = \llbracket A \rrbracket \& \llbracket B \rrbracket \quad \llbracket A \vee B \rrbracket = \varepsilon \quad \llbracket \{\ell : A\} \rrbracket = \{\ell : \llbracket A \rrbracket\} \\
\llbracket \text{sig } \langle T \rangle \rrbracket &= \llbracket T \rrbracket \quad \llbracket \text{sig } (T \Rightarrow \sigma) \rrbracket = \llbracket T \rrbracket \rightarrow \llbracket \text{sig } \sigma \rrbracket \quad \llbracket X_k \rrbracket = \varepsilon \quad \llbracket \mu T \rrbracket = \varepsilon \quad \llbracket \alpha_n \rrbracket = \alpha_n
\end{aligned}$$

Import packages $\text{wire}(\Gamma, c_1, D), \quad \text{WireOk } \Gamma \text{ } \Gamma_1 \text{ } D$

$$\text{wire}(\Gamma, c_1, \{\ell : A\}) = \{\ell = c_1.\ell\} \quad \text{wire}(\Gamma, c_1, D \& \{\ell : A\}) = \text{wire}(\Gamma, c_1, D) \# ((? : \Gamma) \triangleright \{\ell = c_1.\ell\}) \quad \text{wire}(\Gamma, c_1, _) = \varepsilon$$

$$\frac{\ell : A \in \Gamma_1}{\text{WireOk } \Gamma \text{ } \Gamma_1 \text{ } \{\ell : A\}} \text{ONE} \quad \frac{\text{WireOk } \Gamma \text{ } \Gamma_1 \text{ } D \quad \ell : A \in \Gamma_1 \quad \llbracket D \rrbracket * \Gamma \quad \llbracket D \rrbracket * \{\ell : \llbracket A \rrbracket\}}{\text{WireOk } \Gamma \text{ } \Gamma_1 \text{ } (D \& \{\ell : A\})} \text{MORE}$$

Elaboration $\Delta; \Gamma \vdash e : A \rightsquigarrow \bar{e}$

$$\begin{array}{c}
\frac{}{\Delta; \Gamma \vdash ? : \Gamma \rightsquigarrow ?} \text{EQUERY} \qquad \frac{}{\Delta; \Gamma \vdash i : \text{Int} \rightsquigarrow i} \text{ELIT} \qquad \frac{}{\Delta; \Gamma \vdash \varepsilon : \varepsilon \rightsquigarrow \varepsilon} \text{EUNIT} \\
\\
\frac{\Delta; \Gamma \vdash e_1 : A \rightarrow B \rightsquigarrow \bar{e}_1 \quad \Delta; \Gamma \vdash e_2 : A \rightsquigarrow \bar{e}_2}{\Delta; \Gamma \vdash e_1 e_2 : B \rightsquigarrow \bar{e}_1 \bar{e}_2} \text{EAPP} \qquad \frac{\Delta; \Gamma \vdash e : A \rightsquigarrow \bar{e} \quad A \ni_k B}{\Delta; \Gamma \vdash e.k : B \rightsquigarrow \bar{e}.k} \text{EPROJ} \\
\\
\frac{\Delta; \Gamma \vdash e_1 : \Gamma' \rightsquigarrow \bar{e}_1 \quad \Delta; \Gamma' \vdash e_2 : A \rightsquigarrow \bar{e}_2}{\Delta; \Gamma \vdash e_1 \triangleright e_2 : A \rightsquigarrow \bar{e}_1 \triangleright \bar{e}_2} \text{EBOX} \\
\\
\frac{\Delta; \Gamma \vdash e_1 : A \rightsquigarrow \bar{e}_1 \quad \Delta; \Gamma \& A \vdash e_2 : B \rightsquigarrow \bar{e}_2 \quad \llbracket A \rrbracket * \llbracket \Gamma \rrbracket \quad \llbracket A \rrbracket * \llbracket B \rrbracket}{\Delta; \Gamma \vdash e_1 \# e_2 : A \& B \rightsquigarrow \bar{e}_1 \# \bar{e}_2} \text{EDMRG} \\
\\
\frac{v_1 \text{ value} \quad v_2 \text{ value} \quad \Delta; \varepsilon \vdash v_1 : A \rightsquigarrow \bar{w}_1 \quad \Delta; \varepsilon \vdash v_2 : B \rightsquigarrow \bar{w}_2 \quad \llbracket A \rrbracket * \llbracket B \rrbracket}{\Delta; \Gamma \vdash v_1 \# v_2 : A \& B \rightsquigarrow \bar{w}_1 \# \bar{w}_2} \text{EVMRG} \\
\\
\frac{\Delta; \Gamma \vdash e_1 : A \rightsquigarrow \bar{e}_1 \quad \Delta; \Gamma \vdash e_2 : B \rightsquigarrow \bar{e}_2 \quad \llbracket A \rrbracket * \llbracket \Gamma \rrbracket \quad \llbracket A \rrbracket * \llbracket B \rrbracket}{\Delta; \Gamma \vdash e_1 _, e_2 : A \& B \rightsquigarrow \bar{e}_1 \# ((? : \llbracket \Gamma \rrbracket) \triangleright \bar{e}_2)} \text{ENMRG} \\
\\
\frac{\Delta; \Gamma \& A \vdash e : B \rightsquigarrow \bar{e} \quad \llbracket \Gamma \rrbracket * \llbracket A \rrbracket}{\Delta; \Gamma \vdash \lambda A.e : A \rightarrow B \rightsquigarrow \lambda(\llbracket A \rrbracket \rightarrow \llbracket B \rrbracket).\bar{e}} \text{ELAM} \qquad \frac{\Delta; \Gamma \vdash e : B \rightsquigarrow \bar{e} \quad \ell : A \in B}{\Delta; \Gamma \vdash e.\ell : A \rightsquigarrow \bar{e}.\ell} \text{ERPROJ} \\
\\
\frac{v_1 \text{ value} \quad \Delta; \varepsilon \vdash v_1 : \Gamma' \rightsquigarrow \bar{v}_1 \quad \Delta; \Gamma' \& A \vdash e_2 : B \rightsquigarrow \bar{e}_2 \quad \llbracket \Gamma' \rrbracket * \llbracket A \rrbracket}{\Delta; \Gamma \vdash \langle v_1, \lambda A.e_2 \rangle : A \rightarrow B \rightsquigarrow \langle \bar{v}_1, \lambda(\llbracket A \rrbracket \rightarrow \llbracket B \rrbracket).\bar{e}_2 \rangle} \text{ECLOS} \\
\\
\frac{\Delta; \Gamma \vdash e : A \rightsquigarrow \bar{e}}{\Delta; \Gamma \vdash \{\ell = e\} : \{\ell : A\} \rightsquigarrow \{\ell = \bar{e}\}} \text{ELREC} \qquad \frac{\Delta; \Gamma \vdash e_1 : A \rightsquigarrow \bar{e}_1 \quad \Delta; \Gamma \& A \vdash e_2 : B \rightsquigarrow \bar{e}_2 \quad \llbracket \Gamma \rrbracket * \llbracket A \rrbracket}{\Delta; \Gamma \vdash \text{let}^A e_1 \text{ in } e_2 : B \rightsquigarrow (\lambda(\llbracket A \rrbracket \rightarrow \llbracket B \rrbracket).\bar{e}_2) \bar{e}_1} \text{ELETB} \\
\\
\frac{\Delta; \Gamma \vdash e_1 : \{\ell : A\} \rightsquigarrow \bar{e}_1 \quad \Delta; \Gamma \& A \vdash e_2 : B \rightsquigarrow \bar{e}_2 \quad \llbracket \Gamma \rrbracket * \llbracket A \rrbracket}{\Delta; \Gamma \vdash \text{open } e_1 \text{ in } e_2 : B \rightsquigarrow (\lambda(\llbracket A \rrbracket \rightarrow \llbracket B \rrbracket).\bar{e}_2) (\bar{e}_1.\ell)} \text{EOPENM} \\
\\
\frac{\Delta; \varepsilon \vdash e : B \rightsquigarrow \bar{e}}{\Delta; \Gamma \vdash \text{struct}^\bullet e : B \rightsquigarrow \varepsilon \triangleright \bar{e}} \text{EMSTRUCT-}\bullet \qquad \frac{\Delta; \Gamma \vdash e : B \rightsquigarrow \bar{e}}{\Delta; \Gamma \vdash \text{struct}^\circ e : B \rightsquigarrow ? \triangleright \bar{e}} \text{EMSTRUCT-}\circ \\
\\
\frac{\Delta; \varepsilon \& A \vdash e : B \rightsquigarrow \bar{e}}{\Delta; \Gamma \vdash \text{functor}^\bullet(A).e : \text{sig}(A \Rightarrow \langle B \rangle) \rightsquigarrow \varepsilon \triangleright \lambda(\llbracket A \rrbracket \rightarrow \llbracket B \rrbracket).\bar{e}} \text{EMFUNCTOR-}\bullet \\
\\
\frac{\Delta; \Gamma \& A \vdash e : B \rightsquigarrow \bar{e} \quad \llbracket \Gamma \rrbracket * \llbracket A \rrbracket}{\Delta; \Gamma \vdash \text{functor}^\circ(A).e : \text{sig}(A \Rightarrow \langle B \rangle) \rightsquigarrow \lambda(\llbracket A \rrbracket \rightarrow \llbracket B \rrbracket).\bar{e}} \text{EMFUNCTOR-}\circ \\
\\
\frac{v_1 \text{ value} \quad \Delta; \varepsilon \vdash v_1 : \Gamma' \rightsquigarrow \bar{v}_1 \quad \Delta; \Gamma' \& A \vdash e_2 : B \rightsquigarrow \bar{e}_2 \quad \llbracket \Gamma' \rrbracket * \llbracket A \rrbracket}{\Delta; \Gamma \vdash \langle v_1, \text{functor}(A).e_2 \rangle : \text{sig}(A \Rightarrow \langle B \rangle) \rightsquigarrow \langle \bar{v}_1, \lambda(\llbracket A \rrbracket \rightarrow \llbracket B \rrbracket).\bar{e}_2 \rangle} \text{EMCLOS} \\
\\
\frac{\Delta; \Gamma \vdash e_1 : \text{sig}(A \Rightarrow \langle B \rangle) \rightsquigarrow \bar{e}_1 \quad \Delta; \Gamma \vdash e_2 : A \rightsquigarrow \bar{e}_2}{\Delta; \Gamma \vdash e_1 \bullet e_2 : B \rightsquigarrow \bar{e}_1 \bar{e}_2} \text{EMAPP} \\
\\
\frac{\Delta; \Gamma \vdash e_1 : \Gamma_1 \rightsquigarrow \bar{e}_1 \quad \Delta; \Gamma \vdash e_2 : \text{sig}(\{\ell : A\} \Rightarrow \langle B \rangle) \rightsquigarrow \bar{e}_2 \quad \ell : A \in \Gamma_1 \quad \llbracket \Gamma_1 \rrbracket * \llbracket \Gamma \rrbracket \quad \llbracket \Gamma_1 \rrbracket * \llbracket B \rrbracket}{\Delta; \Gamma \vdash e_1 \ltimes e_2 : \Gamma_1 \& B \rightsquigarrow \bar{e}_1 \# ((? : \llbracket \Gamma \rrbracket) \triangleright (\bar{e}_2 \{\ell = \bar{e}_1.\ell\}))} \text{EMLINK} \\
\\
\frac{\Delta; \Gamma \vdash e_1 : \Gamma_1 \rightsquigarrow \bar{e}_1 \quad \Delta; \Gamma \vdash e_2 : \text{sig}(D \Rightarrow \langle B \rangle) \rightsquigarrow \bar{e}_2 \quad \text{WireOk } \llbracket \Gamma \rrbracket \Gamma_1 D \quad \llbracket \Gamma_1 \rrbracket * \llbracket \Gamma \rrbracket \quad \llbracket \Gamma_1 \rrbracket * \llbracket B \rrbracket}{\Delta; \Gamma \vdash e_1 \ltimes^* e_2 : \Gamma_1 \& B \rightsquigarrow \bar{e}_1 \# ((? : \llbracket \Gamma \rrbracket) \triangleright (\bar{e}_2 \text{wire}(\llbracket \Gamma \rrbracket, \bar{e}_1, D)))} \text{EMLINKN} \\
\\
\frac{\Delta n = R \quad v \text{ value} \quad \overset{7}{\Delta}; \varepsilon \vdash v : R \rightsquigarrow \bar{w}}{\Delta; \Gamma \vdash \langle n \rangle v : \alpha_n \rightsquigarrow \langle n \rangle \bar{w}} \text{EWRAP}
\end{array}$$

2.3 Semantics

Value lookups, selection, packages $v \ni_k v', \quad v.\ell \rightsquigarrow v', \quad v \gg_D p$

$$\begin{array}{c}
\frac{}{v_1 \# v_2 \ni_0 v_2} \text{DMRG-ZERO} \qquad \frac{v_1 \ni_k v'}{v_1 \# v_2 \ni_{k+1} v'} \text{DMRG-SUCC} \qquad \frac{}{v_1, v_2 \ni_0 v_2} \text{NMRG-ZERO} \\
\\
\frac{v_1 \ni_k v'}{v_1, v_2 \ni_{k+1} v'} \text{NMRG-SUCC} \qquad \frac{}{\{\ell = v\}.\ell \rightsquigarrow v} \text{RCD} \qquad \frac{v_1.\ell \rightsquigarrow v'}{v_1 \# v_2.\ell \rightsquigarrow v'} \text{DMRG-LEFT} \\
\\
\frac{v_2.\ell \rightsquigarrow v'}{v_1 \# v_2.\ell \rightsquigarrow v'} \text{DMRG-RIGHT} \qquad \frac{v_1.\ell \rightsquigarrow v'}{v_1, v_2.\ell \rightsquigarrow v'} \text{NMRG-LEFT} \qquad \frac{v_2.\ell \rightsquigarrow v'}{v_1, v_2.\ell \rightsquigarrow v'} \text{NMRG-RIGHT} \\
\\
\frac{v.\ell \rightsquigarrow v_\ell}{v \gg_{\{\ell:A\}} \{\ell = v_\ell\}} \text{ONE} \qquad \frac{v \gg_D p \quad v.\ell \rightsquigarrow v_\ell}{v \gg_{D \& \{\ell:A\}} p \# \{\ell = v_\ell\}} \text{MORE}
\end{array}$$

Sealing coercions $v \Downarrow_S^{\text{seal } n R} w, \quad v \Downarrow_S^{\text{unseal } n R} w \quad (\text{pxe}(c) := \varepsilon \# \{\#f = c\}, \quad \text{pxf} := (?).1.\#f)$

$$\begin{array}{c}
\frac{}{v \Downarrow_{\alpha_n}^{\text{seal } n R} \langle n \rangle v} \text{BRAND-EQ} \qquad \frac{m \neq n}{v \Downarrow_{\alpha_m}^{\text{seal } n R} v} \text{BRAND-NE} \qquad \frac{}{i \Downarrow_{\text{Int}}^{\text{seal } n R} i} \text{INT} \qquad \frac{}{v \Downarrow_{\varepsilon}^{\text{seal } n R} \varepsilon} \text{TOP} \\
\\
\frac{v_1 \Downarrow_A^{\text{seal } n R} w_1 \quad v_2 \Downarrow_B^{\text{seal } n R} w_2}{v_1 \# v_2 \Downarrow_{A \& B}^{\text{seal } n R} w_1 \# w_2} \text{AND} \qquad \frac{v \Downarrow_A^{\text{seal } n R} w}{\{\ell = v\} \Downarrow_{\{\ell:A\}}^{\text{seal } n R} \{\ell = w\}} \text{RCD} \\
\\
\frac{}{c \Downarrow_{A \rightarrow B}^{\text{seal } n R} \langle \text{pxe}(c), \lambda A. \text{seal } n R B (\text{pxf} (\text{unseal } n R A ?).0) \rangle} \text{ARR} \\
\\
\frac{}{\langle n \rangle v \Downarrow_{\alpha_n}^{\text{unseal } n R} v} \text{BRAND-EQ} \qquad \frac{m \neq n}{v \Downarrow_{\alpha_m}^{\text{unseal } n R} v} \text{BRAND-NE} \qquad \frac{}{i \Downarrow_{\text{Int}}^{\text{unseal } n R} i} \text{INT} \qquad \frac{}{v \Downarrow_{\varepsilon}^{\text{unseal } n R} \varepsilon} \text{TOP} \\
\\
\frac{v_1 \Downarrow_A^{\text{unseal } n R} w_1 \quad v_2 \Downarrow_B^{\text{unseal } n R} w_2}{v_1 \# v_2 \Downarrow_{A \& B}^{\text{unseal } n R} w_1 \# w_2} \text{AND} \qquad \frac{v \Downarrow_A^{\text{unseal } n R} w}{\{\ell = v\} \Downarrow_{\{\ell:A\}}^{\text{unseal } n R} \{\ell = w\}} \text{RCD} \\
\\
\frac{}{c \Downarrow_{A \rightarrow B}^{\text{unseal } n R} \langle \text{pxe}(c), \lambda A[n:=R]. \text{unseal } n R B (\text{pxf} (\text{seal } n R A ?).0) \rangle} \text{ARR}
\end{array}$$

Big-step evaluation $\rho \vdash e \Downarrow v$ (all rules carry ρ value; value premises shown)

$$\begin{array}{c}
\frac{}{\rho \vdash ? \Downarrow \rho} \text{QUERY} \quad \frac{}{\rho \vdash i \Downarrow i} \text{LIT} \quad \frac{}{\rho \vdash \varepsilon \Downarrow \varepsilon} \text{UNIT} \quad \frac{v \text{ value}}{\rho \vdash \langle v, \lambda A.e \rangle \Downarrow \langle v, \lambda A.e \rangle} \text{CLOS-VAL} \\
\\
\frac{v \text{ value}}{\rho \vdash \langle v, \text{functor}(A).e \rangle \Downarrow \langle v, \text{functor}(A).e \rangle} \text{MCLOS-VAL} \quad \frac{\rho \vdash e \Downarrow v \quad v \exists_k v'}{\rho \vdash e.k \Downarrow v'} \text{PROJ} \quad \frac{}{\rho \vdash \lambda A.e \Downarrow \langle \rho, \lambda A.e \rangle} \text{LAM} \\
\\
\frac{\rho \vdash e_1 \Downarrow v_1 \quad v_1 \vdash e_2 \Downarrow v}{\rho \vdash e_1 \triangleright e_2 \Downarrow v} \text{BOX} \quad \frac{\rho \vdash e_1 \Downarrow \langle v_1, \lambda A.e \rangle \quad \rho \vdash e_2 \Downarrow v_2 \quad v_1 \# v_2 \vdash e \Downarrow v}{\rho \vdash e_1 e_2 \Downarrow v} \text{APP-CLOS} \\
\\
\frac{\rho \vdash e_1 \Downarrow \langle v_1, \mu\lambda(A \rightarrow B).e \rangle \quad \rho \vdash e_2 \Downarrow v_2 \quad (v_1 \# \langle v_1, \mu\lambda(A \rightarrow B).e \rangle) \# v_2 \vdash e \Downarrow v}{\rho \vdash e_1 e_2 \Downarrow v} \text{APP-FCLOS} \\
\\
\frac{\rho \vdash e_1 \Downarrow \langle v_1, \text{functor}(A).e \rangle \quad \rho \vdash e_2 \Downarrow v_2 \quad v_1 \# v_2 \vdash e \Downarrow v}{\rho \vdash e_1 \bullet e_2 \Downarrow v} \text{APP-MCLOS} \\
\\
\frac{\rho \vdash e_1 \Downarrow v_1 \quad \rho \# v_1 \vdash e_2 \Downarrow v_2}{\rho \vdash e_1 \# e_2 \Downarrow v_1 \# v_2} \text{DMRG} \quad \frac{\rho \vdash e_1 \Downarrow v_1 \quad \rho \vdash e_2 \Downarrow v_2}{\rho \vdash e_1 ,, e_2 \Downarrow v_1 \# v_2} \text{NMRG} \quad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \{\ell = e\} \Downarrow \{\ell = v\}} \text{LREC} \\
\\
\frac{\rho \vdash e \Downarrow v \quad v.\ell \rightsquigarrow v'}{\rho \vdash e.\ell \Downarrow v'} \text{RPROJ} \quad \frac{\rho \vdash e_1 \Downarrow v_1 \quad \rho \# v_1 \vdash e_2 \Downarrow v}{\rho \vdash \text{let}^A e_1 \text{ in } e_2 \Downarrow v} \text{LETB} \\
\\
\frac{\rho \vdash e_1 \Downarrow \{\ell = v'\} \quad \rho \# v' \vdash e_2 \Downarrow v}{\rho \vdash \text{open } e_1 \text{ in } e_2 \Downarrow v} \text{OPENM} \quad \frac{\varepsilon \vdash e \Downarrow v}{\rho \vdash \text{struct}^\bullet e \Downarrow v} \text{MSTRUCT-}\bullet \quad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \text{struct}^\circ e \Downarrow v} \text{MSTRUCT-}\circ \\
\\
\frac{}{\rho \vdash \text{functor}^\bullet(A).e \Downarrow \langle \varepsilon, \text{functor}(A).e \rangle} \text{MFUNCTOR-}\bullet \quad \frac{}{\rho \vdash \text{functor}^\circ(A).e \Downarrow \langle \rho, \text{functor}(A).e \rangle} \text{MFUNCTOR-}\circ \\
\\
\frac{\rho \vdash e_1 \Downarrow v_1 \quad \rho \vdash e_2 \Downarrow \langle v_2, \text{functor}(\{\ell : A\}).e \rangle \quad v_1.\ell \rightsquigarrow v_\ell \quad v_2 \# \{\ell = v_\ell\} \vdash e \Downarrow v_3}{\rho \vdash e_1 \ltimes e_2 \Downarrow v_1 \# v_3} \text{MLINK} \\
\\
\frac{\rho \vdash e_1 \Downarrow v_1 \quad \rho \vdash e_2 \Downarrow \langle v_2, \text{functor}(D).e \rangle \quad v_1 \gg_D p \quad v_2 \# p \vdash e \Downarrow v_3}{\rho \vdash e_1 \ltimes^* e_2 \Downarrow v_1 \# v_3} \text{MLINKN} \quad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \text{inl}^B e \Downarrow \text{inl}^B v} \text{INL} \\
\\
\frac{\rho \vdash e \Downarrow v}{\rho \vdash \text{inr}^A e \Downarrow \text{inr}^A v} \text{INR} \quad \frac{\rho \vdash e \Downarrow \text{inl}^B v_1 \quad \rho \# v_1 \vdash e_1 \Downarrow v}{\rho \vdash \text{case } e \text{ of } e_1 \mid e_2 \Downarrow v} \text{CASE-INL} \\
\\
\frac{\rho \vdash e \Downarrow \text{inr}^A v_1 \quad \rho \# v_1 \vdash e_2 \Downarrow v}{\rho \vdash \text{case } e \text{ of } e_1 \mid e_2 \Downarrow v} \text{CASE-INR} \quad \frac{v \text{ value}}{\rho \vdash \langle v, \mu\lambda(A \rightarrow B).e \rangle \Downarrow \langle v, \mu\lambda(A \rightarrow B).e \rangle} \text{FCLOS-VAL} \\
\\
\frac{}{\rho \vdash \mu\lambda(A \rightarrow B).e \Downarrow \langle \rho, \mu\lambda(A \rightarrow B).e \rangle} \text{FLAM} \quad \frac{\rho \vdash e \Downarrow v}{\rho \vdash \text{fold}^T e \Downarrow \text{fold}^T v} \text{FOLD} \quad \frac{\rho \vdash e \Downarrow \text{fold}^T v}{\rho \vdash \text{unfold } e \Downarrow v} \text{UNFOLD} \\
\\
\frac{\rho \vdash e \Downarrow v}{\rho \vdash \langle n \rangle e \Downarrow \langle n \rangle v} \text{WRAP} \quad \frac{\rho \vdash e \Downarrow v \quad v \Downarrow_S^{\text{seal } n R} w}{\rho \vdash \text{seal } n R S e \Downarrow w} \text{MSEAL} \quad \frac{\rho \vdash e \Downarrow v \quad v \Downarrow_S^{\text{unseal } n R} w}{\rho \vdash \text{unseal } n R S e \Downarrow w} \text{MUNSEAL}
\end{array}$$

2.4 Meta-theory

Definition 2.1 (EVal $\Delta \vdash v \sim_A \bar{v}$).

$$\begin{array}{c}
\frac{}{\Delta \vdash i \sim_{\text{Int}} i}^{\text{LIT}} \quad \frac{}{\Delta \vdash \varepsilon \sim_{\varepsilon} \varepsilon}^{\text{UNIT}} \quad \frac{\Delta \vdash v_1 \sim_A \bar{w}_1 \quad \Delta \vdash v_2 \sim_B \bar{w}_2 \quad \llbracket A \rrbracket * \llbracket B \rrbracket}{\Delta \vdash v_1 \# v_2 \sim_{A \& B} \bar{w}_1 \# \bar{w}_2}^{\text{MRG}} \\
\\
\frac{\Delta \vdash v \sim_A \bar{w}}{\Delta \vdash \{\ell = v\} \sim_{\{\ell:A\}} \{\ell = \bar{w}\}}^{\text{RCD}} \quad \frac{\rho \text{ value} \quad \Delta \vdash \rho \sim_{\Gamma'} \bar{\rho} \quad \Delta; \Gamma' \& A \vdash e : B \rightsquigarrow \bar{e} \quad \llbracket \Gamma' \rrbracket * \llbracket A \rrbracket}{\Delta \vdash \langle \rho, \lambda A. e \rangle \sim_{A \rightarrow B} \langle \bar{\rho}, \lambda (\llbracket A \rrbracket \rightarrow \llbracket B \rrbracket). \bar{e} \rangle}^{\text{CLOS}} \\
\\
\frac{\rho \text{ value} \quad \Delta \vdash \rho \sim_{\Gamma'} \bar{\rho} \quad \Delta; \Gamma' \& A \vdash e : B \rightsquigarrow \bar{e} \quad \llbracket \Gamma' \rrbracket * \llbracket A \rrbracket}{\Delta \vdash \langle \rho, \text{functor}(A).e \rangle \sim_{\text{sig}(A \Rightarrow \langle B \rangle)} \langle \bar{\rho}, \lambda (\llbracket A \rrbracket \rightarrow \llbracket B \rrbracket). \bar{e} \rangle}^{\text{MCLOS}} \\
\\
\frac{\llbracket A \rrbracket \uparrow \quad v \text{ value} \quad \Delta; \varepsilon \vdash v : A \rightsquigarrow \bar{w}}{\Delta \vdash v \sim_A \llbracket A \rrbracket^\uparrow}^{\text{GEN}} \quad \frac{\Delta n = R \quad \Delta \vdash v \sim_R \bar{w}}{\Delta \vdash \langle n \rangle v \sim_{\alpha_n} \langle n \rangle \bar{w}}^{\text{WRAP}}
\end{array}$$

Theorem 2.2 (seal_type_preservation).

$$\Delta; \Gamma \vdash e : A \rightsquigarrow \bar{e} \implies \llbracket \Delta \rrbracket; \llbracket \Gamma \rrbracket \vdash \bar{e} : \llbracket A \rrbracket$$

Theorem 2.3 (elabSeal_weaken_store).

$$\Delta \sqsubseteq \Delta' \wedge \Delta; \Gamma \vdash e : A \rightsquigarrow \bar{e} \implies \Delta'; \Gamma \vdash e : A \rightsquigarrow \bar{e}$$

Theorem 2.4 (seal_semantic_preservation).

$$\rho \vdash e \Downarrow v \wedge \Delta; \Gamma \vdash e : A \rightsquigarrow \bar{e} \wedge \rho \text{ value} \wedge \Delta \vdash \rho \sim_{\Gamma} \bar{\rho} \implies \exists \bar{v}. \bar{\rho} \vdash \bar{e} \hookrightarrow^* \bar{v} \wedge \Delta \vdash v \sim_A \bar{v}$$

Theorem 2.5 (seal_whole_program_correctness).

$$\Delta; \varepsilon \vdash e : A \rightsquigarrow \bar{e} \wedge \varepsilon \vdash e \Downarrow v \implies \exists \bar{v}. \varepsilon \vdash \bar{e} \hookrightarrow^* \bar{v} \wedge \Delta \vdash v \sim_A \bar{v}$$

Theorem 2.6 (seal_separate_compilation).

$$\begin{aligned}
& \Delta; \Gamma \vdash e_1 : \Gamma_1 \rightsquigarrow \bar{e}_1 \wedge \Delta; \Gamma \vdash e_2 : \text{sig}(\{\ell : A\} \Rightarrow \langle B \rangle) \rightsquigarrow \bar{e}_2 \wedge \ell : A \in \Gamma_1 \wedge \llbracket \Gamma_1 \rrbracket * \llbracket \Gamma \rrbracket \wedge \llbracket \Gamma_1 \rrbracket * \llbracket B \rrbracket \wedge \\
& \rho \vdash e_1 \times e_2 \Downarrow v \wedge \rho \text{ value} \wedge \Delta \vdash \rho \sim_{\Gamma} \bar{\rho} \\
& \implies \exists \bar{v}. \bar{\rho} \vdash \bar{e}_1 \# ((? : \llbracket \Gamma \rrbracket) \triangleright (\bar{e}_2 \{\ell = \bar{e}_1.\ell\})) \hookrightarrow^* \bar{v} \wedge \Delta \vdash v \sim_{\Gamma_1 \& B} \bar{v}
\end{aligned}$$

Theorem 2.7 (seal_separate_compilation_sealed).

$$\begin{aligned}
& \Delta n = R \wedge \text{NoRes } \llbracket R \rrbracket \wedge \text{WfSig } n \llbracket R \rrbracket \llbracket \Gamma_1 \rrbracket \wedge \Delta; \Gamma \vdash e_1 : \Gamma_1[n:=R] \rightsquigarrow \bar{e}_1 \wedge \Delta_c \sqsubseteq \Delta \wedge \\
& \Delta_c; \Gamma \vdash e_2 : \text{sig}(\{\ell : A\} \Rightarrow \langle B \rangle) \rightsquigarrow \bar{e}_2 \wedge \ell : A \in \Gamma_1 \wedge \llbracket \Gamma_1 \rrbracket * \llbracket \Gamma \rrbracket \wedge \llbracket \Gamma_1 \rrbracket * \llbracket B \rrbracket \wedge \\
& \rho \vdash (\text{seal } n \ R \ \Gamma_1 \ e_1) \times e_2 \Downarrow v \wedge \rho \text{ value} \wedge \Delta \vdash \rho \sim_{\Gamma} \bar{\rho} \\
& \implies \exists \bar{v}. \bar{\rho} \vdash \bar{s} \# ((? : \llbracket \Gamma \rrbracket) \triangleright (\bar{e}_2 \{\ell = \bar{s}.\ell\})) \hookrightarrow^* \bar{v} \wedge \Delta \vdash v \sim_{\Gamma_1 \& B} \bar{v} \quad \text{where } \bar{s} := \text{seal } n \ \llbracket R \rrbracket \llbracket \Gamma_1 \rrbracket \bar{e}_1
\end{aligned}$$

Theorem 2.8 (source_representation_independence).

$$\begin{aligned}
& \text{OpenBrand } \llbracket \Delta_1 \rrbracket \llbracket \Delta_2 \rrbracket \eta \ n \ \llbracket R_1 \rrbracket \llbracket R_2 \rrbracket \wedge \Delta_1 n = R_1 \wedge \Delta_2 n = R_2 \wedge \\
& \text{WfSig } n \llbracket R_1 \rrbracket \llbracket S \rrbracket \wedge \text{WfSig } n \llbracket R_2 \rrbracket \llbracket S \rrbracket \wedge \Delta_1; \varepsilon \vdash p_1 : S[n:=R_1] \rightsquigarrow \bar{p}_1 \wedge \Delta_2; \varepsilon \vdash p_2 : S[n:=R_2] \rightsquigarrow \bar{p}_2 \wedge \\
& \mathcal{V}[\llbracket S \rrbracket]^{(n, \llbracket R_1 \rrbracket, \llbracket R_2 \rrbracket)}(\bar{p}_1, \bar{p}_2) \wedge \emptyset; S \vdash c : \text{Int} \rightsquigarrow \bar{c} \wedge \\
& \varepsilon \vdash (\text{seal } n \ R_1 \ S \ p_1) \triangleright c \Downarrow v_1 \wedge \varepsilon \vdash (\text{seal } n \ R_2 \ S \ p_2) \triangleright c \Downarrow v_2 \\
& \implies \exists i. v_1 = i \wedge v_2 = i
\end{aligned}$$